Operational semantics for Prolog with Cut in Rocq and its application to determinacy analysis

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Abstract

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2012 ACM Subject Classification Replace ccsdesc macro with valid one

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1 Introduction

Elpi is a dialect of λprolog (see [14, 15, 7, 12]) used as an extension language for the Rocq prover (formerly the Coq proof assistant). Elpi has become an important infrastructure component: several projects and libraries depend on it [13, 3, 4, 19, 8, 9]. Examples include the Hierarchy-Builder library-structuring tool [5] and Derive [17, 18, 11], a program-and-proof synthesis framework with industrial applications at SkyLabs AI.

Starting with version 3, Elpi gained a static analysis for determinacy [10] to help users tame backtracking. Rocq users are familiar with functional programming but not necessarily with logic programming and uncontrolled backtracking is a common source of inefficiency and makes debugging harder. The determinacy checkers identifies predicates that behave like functions, i.e., predicates that commit to their first solution and leave no *choice points* (places where backtracking could resume).

This paper reports our first steps towards a mechanization, in the Rocq prover, of the determinacy analysis from [10]. We focus on the control operator *cut*, which is useful to restrict backtracking but makes the semantic depart from a pure logical reading.

We formalize two operational semantics for Prolog with cut. The first is a stack-based semantics that closely models Elpi's implementation and is similar to the semantics mechanized by Pusch in Isabelle/HOL [16] and to the model of Debray and Mishra [6, Sec. 4.3]. This stack-based semantics is a good starting point to study further optimizations used by standard Prolog abstract machines [20, 1], but it makes reasoning about the scope of *cut* difficult. To address that limitation we introduce a tree-based semantics in which the branches pruned by *cut* are explicit and we prove the two semantics equivalent. Using the

[1] Optional footnote, e.g. to mark corresponding author

```
Inductive P := IP of nat. Inductive D := ID of nat. Inductive V := IV of nat.

Inductive Tm :=
  | Tm_P of P      | Tm_D    of D
  | Tm_V of V      | Tm_App  of Tm & Tm.
```

Figure 1 Tm types

```
Record Unif := {
  unify : Tm → Tm → Σ → option Σ;
  matching : Tm → Tm → Σ → option Σ;
}.
```

Figure 2 Unif type

tree-based semantics we then show that if every rule of a predicate passes the determinacy analysis, the call to a deterministic predicate does not leave any choice points.

2 Common code: the language

put unif and program in variable hides from types

Before going to the two semantcis, we show the piece of data structure that are shared by the them. The smallest unit of code that we can use in the langauge is an atom. The atom inductive (see Type 1) is either a cut or a call. A call carries a term (see Figure 1). A term (Tm) is either a predicate, a datum, a variable or the binary application of two terms.

```
Inductive A := cut | call : Tm → A.                                    (1)
Record R := mkR { head : Tm; premises : list A }.                      (2)
Record ℙ := { rules : seq R; sig : sigT }.                             (3)
Definition Σ := {fmap V → Tm}.                                         (4)
Definition bc : ℙ → ℱᵥ → Tm → Σ → ℱᵥ * seq (Σ * seq A) :=              (5)
```

A rule (see Type 2) has a head of type term and a list of premises, that are a list of atoms. A program (see Type 3) is made by its rules and the signatures of predicates, `sigT` is a mapping from predicates to their signatures, signatures represent the defintion of type from the simply typed lambda calculus, i.e. it is either a base type or an arrow for term application. We decorate arrows to know the mode (input or output) of the application argument.

A substitution (see Type 4) is a mapping from variables to terms. It is the output of a successful query. The empty substitution is noted ϵ.

The backchain function (bc, see Type 5) takes: a program P; a set fv of free variables ($\mathcal{F}_\nu$); a query q; and the substitution σ. A backchain is performed when the current atom is a `call`. In this case, it start by recursively *dereferencing* the variables in σ, i.e. replacing all the variables in the query by their term mapping in σ.

TODO: free variables and program freshness

Then it takes the head of the term, which should be rigid, i.e. a predicate p, and look for the signature s of p in the program. Then, for each rule r in P, it unifies the head of r with q. Unification is possible thanks to the unifcator U, whose type is shown in Figure 2. U explains how to unify terms up to standard unifcation (for output terms) or matching (for input terms). The function *unif* (resp. *matching*) unifies (resp. matches) the two terms under σ, it return an extension of σ if the two terms can be unified (resp. matched), and None otherwise. The result of a backchain is made by filtering the rules in P that can be used on the given query. In particular, for each rule filtered rules r, it returnes a pair containing the extension of σ making the head of r and q unify and the list of premises of r.

```
f 1 2.   f 2 3.    r 0 0.   r 0 1.   r 0 2.
g X Z :- r X Y, f Y Z, !.  % r1
g X Z :- f X Y, f Y Z.     % r2
```

Figure 3 Small ELPI program example

```
Inductive tree :=
  | KO | OK | TA of A
  | Or  of option tree & Σ & tree
  | And of tree & seq A & tree.
```

Figure 4 Tree inductive

2.1 The cut operator

ELPI is a dialect of λPROLOG with the hard cut operator and, in the language we have shown before, *cut* is an atom. The semantics of the cut operator adopted in the ELPI language corresponds to the *hard cut* operator of standard SWI-PROLOG. This operator has two primary purposes. First, it eliminates all alternatives that are created either simultaneously with, or after, the introduction of the cut into the execution state.

se metti r1 = g A B :- f A B. allora g e f sono funzioni, e puoi spiegare anche l'idea del detcheck qui

The ELPI program we will use as an example in the following sections is shown in Figure 3.

To illustrate the high-level description of *cut*, consider the program P shown in Figure 3 and the query $q =$ `g 0 R`. Both rules for `g` can be used on the query q. Backchaining q in P from the empty (ϵ) subsitution returns the list $[(\sigma_1, [\texttt{r X Y, f Y Z, !}]), (\sigma_2, [\texttt{f X Y, f Y Z}])]$ with σ_1 and σ_1 both equal to $\{\texttt{X} \mapsto 0, \texttt{Z} \mapsto \texttt{R}\}$. The two elements of the list are generated respectively from the rules r_1 and r_2.

The next step of the interpretation corresponds to the backchaining of `r X Y` from σ_1. This returns $[(\sigma_3, []), (\sigma_4, []), (\sigma_5, [])]$ with $\sigma_3 \coloneqq \sigma_1 + \{\texttt{Y} \mapsto 0\}$, $\sigma_4 \coloneqq \sigma_1 + \{\texttt{Y} \mapsto 1\}$ and $\sigma_5 \coloneqq \sigma_1 + \{\texttt{Y} \mapsto 2\}$. The three lists of premises are empty since all rules for `r` have no premises.

Backchaining `f Y Z` in σ_3 gives no solution, therefore, by backtracking, we backchain `f Y Z` in σ_4. This gives a solution, namely $\sigma_6 \coloneqq \sigma_4 + \{\texttt{R} \mapsto 2\}$.

We now reach the cut operator and can see its double side behavior. In the current execution state, we have still two unexplored alternatives: we can backtrack on the third solution for `r` that assign `Y` to `2` and on the rules r_2. We can note, however, the first alternative have been generated after the cut introduction, and the second alternative have been generated at the same moment of the cut. Therefore both are cut away leaving no choice point. The final complete solution is: $\{\texttt{X} \mapsto 0, \texttt{Z} \mapsto \texttt{R}, \texttt{Y} \mapsto 1, \texttt{R} \mapsto 2\}$, where `X`, `Y` and `Z` can be ignored since they are the result internal unification.

3 Semantics intro

We propose two operational semantics for a logic program with cut. The two semantics are based on different syntaxes, the first syntax (called tree) exploits a tree-like structure and is ideal both to have a graphical view of its evolution while the tree is being intepreted and to prove lemmas over it. The second syntax, called elpi, is the ELPI's syntax and has the advantage of reducing the computational cost of cutting and backtracking alternatives by using shared pointers. We aim to prove the equivalence of the two semantics together with some interesting lemmas of the cut behavior.

4 Tree semantics

The tree state The tree inductive, shown in Figure 4. We distinguish 5 main cases: *KO*, *OK*, and are special meta-symbols representing, respectively, the basic failed and a successful

```
Fixpoint get_end s A : Σ * tree:=
  match A with
  | TA _ | KO | OK ⇒ (s, A)
  | Or None s₁ B ⇒ get_end s₁ B
  | Or (Some A) _ _ ⇒ get_end s A
  | And A _ B ⇒
    let (s', pA) := get_end s A in
    if pA == OK then get_end s' B
    else (s', pA)
  end.
```

(a) Defintion of *get_end*

```
Definition get_subst s A := (get_end s A).1.

Definition path_end A := (get_end ϵ A).2.

Definition success A := path_end A == OK.

Definition failed A := path_end A == KO.

Definition path_atom A :=
    if path_end A is TA _ then true
    else false.
```

(b) Functions from *get_end*

Figure 5 *get_end* and derives functions

trees. These symbols are considered meta because they are internal intermediate symbols used to give structure to the tree.

The *TA* constructor (acronym for tree-atom) is the constructor of atoms in the tree. They are either the cut operator or a call, in these cases, the interpreter should evaluate the atom by either cutting the subtrees in the scope of the cut or backchaining the call in the program.

The two recursive cases of a tree are the *Or* and *And* nodes. The *Or* node $A \vee B_\sigma$ denotes a disjunction between two trees A and B. The first branch is optional, if absent it represents a dead tree, i.e. a tree that has been entirely explored. The second branch is annotated with a suspended substitution σ so that, upon backtracking to B, σ is used as the initial substitution for the execution of B.

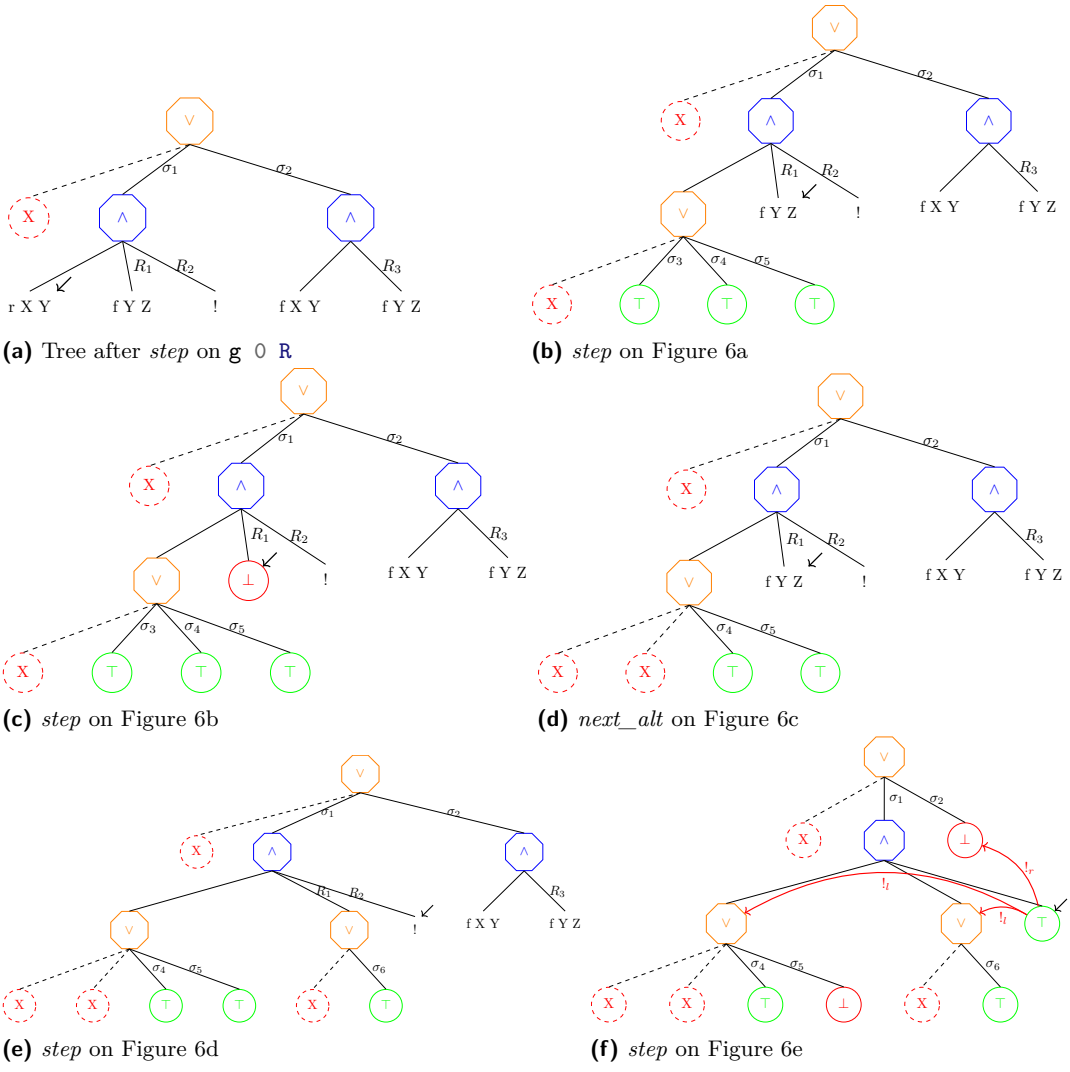
The *And* node $A \wedge_{B_0} B$ represents a conjunction of two trees A and B. We call B_0 the reset point for B; it is used to restore the state of B to its initial form if a backtracking operation occurs on A. An example of this behavior is shown in Section 4.2

A graphical representation of a tree is shown in Figure 6a. To make the graph more compact, the *And* and *Or* nodes are n-ary rather than binary, and the children of these nodes associate to the right. The *KO* node acts as the neutral elements in the *Or* list, while *OK* is the neutral element of the *And* list.

The tree interpreter The interpretation of a tree is performed by two main routines: *step* and *next_alt* that traverse the tree depth-first, left-to-right. Then, then *runT* inductive makes the transitive closure of step *step* and *next_alt*: it iterates the calls to its these auxiliary functions as much as needed. In Types 7–9 we give the types contrats of these symbols where $\mathcal{F}_\nu$ is a set of variable names.

```
Inductive tag := Expanded | CutBrothers | Failed | Success.                    (6)
Definition step : ℙ → Fν → Σ → tree → (Fν * tag * tree) :=                     (7)
Definition next_alt : 𝔹 → tree → option tree :=                               (8)
Inductive runT (p : ℙ): Fν → Σ → tree →
                  Σ → option tree → Prop :=                                   (9)
```

The tree interpreter, as in prolog, explores the tree in DFS strategy, to discover the substitution and the leaf of the tree that should be interpreted. These two pieces of information can be retrieved for a substitution s and a tree A thanks to the *get_end* routine, shown in Figure 5a If the tree is already a leaf, *get_end* returns its inputs. Otherwise, if the tree is a disjunction, the path continues on the left subtree, if it exists, otherwise *get_end* contues on the ths using the substitution stored in the *Or* node. In the case of a conjunction,



(a) Tree after *step* on `g 0 R`

(b) *step* on Figure 6a

(c) *step* on Figure 6b

(d) *next_alt* on Figure 6c

(e) *step* on Figure 6d

(f) *step* on Figure 6e

Figure 6 Some tree representations[2]

if *get_end* of the lhs returns the tree *OK*, then the *get_end* continues in the rhs, otherwise we return the result of the lhs. We derive some useful definition from *get_end*. They are listed in Figure 5b.

We have pointed by a black arraw the *path_end* of all the trees in Figure 6.

4.1 The *step* procedure

The *step* procedure takes as input a program, a set of variables, a substitution, and a tree, and returns an updated set of variables, a *tag*, and an updated tree.

The set of variables *fv* given in input to step are used and updated when a backchaining operation takes place. A call to *step* produces a good result if *fv* contains all the variables

[2] The substitution $\sigma_1 \ldots \sigma_6$ are from Section 2.1, R_1, R_2 and R_3 are the reset points and correspond respectively to the list of atoms [`f Y Z, !`], [`!`] and [`f Y Z`]

not true in figure:fix + also fix in Section 2.1

in the tree. This guarantees that any call to *bc* returns a list of atoms whose variables are disjoint from the variables of the current working tree.

The *tag* (see Type 6) indicates the kind of an internal tree step: `CutBrothers` denotes the interpretation of a superficial cut, i.e., a cut whose parent nodes are all *And*-nodes. `Expanded` denotes the interpretation of non-superficial cuts or predicate calls. `Failure` and `Success` are returned for, respectively, *failed* and *success* trees.

The step procedure is intended to evaluate atoms, that is, it leaves unchanged the tree iff its *path_end* is not an atom, i.e. if it is a success- or a failed-tree.

```
Lemma succ_step_iff u p fv s A: success A ↔ step u p fv s A = (fv, Success, A).   (1)
Lemma fail_step_iff u p fv s A: failed A ↔ step u p fv s A = (fv, Failed, A).     (2)
```

The interpretation of a call to a term c stars by calling the *bc* function on c. The result of the backchain is a list l. If l is empty then the current call is replaced by the *KO* node, otherwise it is replaced by a right skewed tree made by *Or* inner-nodes and each leave correspod to a list of atom e in l. If e is empty then it is mapped to the *OK* tree, otherwise it is mapped to a right-skewed tree made by *And* inner-nodes.

In Figure 6a, we provide a clear example of a call to *step* on the query `q 0 R` on the program P in Figure 3. Let c_0, c_1 and c_2 be the childreen of the root. c_0, by construction is the *None* tree, while c_1 and c_2 correspod respectively to the premises of rules r_1 and r_2 in P. We need the first *None* subtree so that we can attach a substituion to c_1. In particular, we note that the c_1 (resp. c_2) are related to the substitution σ_1 (resp. σ_2). The value of σ_1 and σ_2 are the same as in Section 2.1. The reset points in the tree are marked with the R symbol. $R_1 \coloneqq$ [`f Y Z, !`], $R_2 \coloneqq$ [`!`] and $R_3 \coloneqq$ `f Y Z`, they are essentially the list of atoms that allowed to generate the current subtree. Other example of call to the *step* procedure triggering a backchain are Figures 6b, 6c, and 6e.

The interpretation of a cut has three main impacts: at first the cut is replaced by the *OK* node, then some special subtrees, in the scope of the *Cut*, are cut away: in particular we need to soft-kill the left-siblings of the *Cut* and hard-kill the right-uncles of the the *Cut*.

▶ **Definition 1** (Left-siblings (resp. right-sibling)). *Given a node A, the left-siblings (resp. right-sibling) of A are the list of subtrees sharing the same parent of A and that appear on its left (resp. right).*

▶ **Definition 2** (Right-uncles). *Given a node A, the right-uncles of A are the list of right-sibling of the father of A.*

▶ **Definition 3** (Hard-kill, $!_r$). *Given a tree t, hard-kill replaces the given subtree with the* KO *node*

▶ **Definition 4** (Soft-kill, $!_l$). *Given a successfull tree t, soft-kill replaces with* KO *all subtrees that are not part of the path in t leading to the* OK *node.*

An example of the impact of the cut is show in **??**, the dashed triangles represent generic trees. The step routine interprets the cut since it is the node in its path-end: we pass throgh a and and all trees on the left of the cut are successful. In the example we have 4 arrow tagged with the $!_l$ or $!_r$ symbols. The $!_l$ arrows go left and soft-kill the pointed subtree, it keeps *OK* nodes since they are part of the tree leading to the cut, and replaces the other subtrees with *KO*. The $!_r$ procedure replaces the nodes pointed by the arrows with *KO*.

4.2 The *next_alt* procedure

It is evident that the *step* alone is not sufficient to reproduce entirely the behavior of the full expected prolog interpreter. In particular, we need to bracktrack on failures. Moreover, in case of success, we should return a tree cleaned of its success, this is essential to, non deterministically, find the next solution of a given query. By Lemmas 1 and 2, we know that *step* returns the identity on successful and failed trees. In order to continue the computation on these particular trees, we need the *next_alt* procedure aiming to expecially work with failed and successful trees: and its implementation in **??**.

The *next_alt* procedure takes a boolean and a tree, clean it from failures or success and returns a new tree if this tree still contains a non explored path. The idea behind *next_alt* is to clean recursively every subtree in DFS order if its *path_end* is a failure. Moreover, if the boolean passed to *next_alt* is true, then it erases the first successful path in the tree.

The base cases of *next_alt* are immidiate. The *Or* case is rathere intuitive: if the lhs of the *Or* does not exist we look for the *next_alt* in the rhs. Otherwise, we look for the *next_alt* in the lhs, if this *next_alt* does not exists, we look for the *next_alt* in the rhs.

We want to spend few words about the *And* case, since the reset point $B0$ for B plays an important role. The *next_alt* in an *And* tree should consider two cases: if the lhs succeeds, then the *next_alt* should be retrived in the rhs. If this alternative does not exists it means that the rhs has entirely been explored. We need to erase the success in the lhs and try to find if a non-explored alternative exists. If so, we return a new tree with the new lhs and the rhs is built from the reset point. `big_and` is a trivial function build a right-skewed tree of and nodes where the leaves are the atoms written in the reset point. We need to reuse the reset point since, the step procedure in *And* trees evaluates the rhs of a *And* tree if the lhs succeeds. This evaluation is dependent on the subsitution in the lhs tree. Therefore, if we need to backtrack in the lhs, we need to reset the rhs.

Some interesting property of *next_alt* are shown below and allow to see how *next_alt* complements *step*.

```
Lemma path_atom_next_alt_id b A: path_atom A → next_alt b A = Some A.   (3)
Lemma next_alt_failedF b A A': next_alt b A = Some A' → failed A' = false.   (4)
```

For example, in **??** the step procedure has created a failed tree: its path-end ends in *KO*. The expected behavior of *next_alt* is to take this *KO* node and make it a .... This allows *step* to continue the exploration of the tree. In particular, the path-end of this new tree ends in *OK*. The step leaves the tree unchanged producing the new substitution. This solution however is not unique, we should be able to backtrack on this successful tree. To do so we can call *next_alt* and it will deadify the *OK* node allowing *step* to proceed on r X Z.

subst taken form the or

4.3 The *runT* inductive

TODO: start
DIRE CHE RUN TERMINA
In our implementation, *runT* gives the garantee that the program interpretation terminates, since it always return a substitution, but we can easily make it also accept diverging program by making *runT* return an optional substitution
TODO: end

The inductive procedure *runT* is modeled as a function: it takes as input a program, a set of free variables, an initial substitution σ_0, and a tree t_0, and returns a substitution σ_1 together with an optional updated tree t_1. The substitution σ_1 represents the most-general

unificator that makes the execution of the tree t_0 succeed starting from the initial substitution σ_0, σ_1 is an extension of σ_0. The tree t_1 is the updated tree containing the alternatives that have not yet been explored. If the tree contains no solution, then `None` is returned.

The procedure *runT* is based on three main derivation rules, shown in Figure 7. If the *path_end* of the tree t is a success, the input substitution is returned and the input tree is cleaned of its successful path. If the *path_end* of the tree is an atom, then *step* is invoked to evaluate this atom, and *runT* is recursively called on the new tree. Finally, if the *path_end* of the tree is a failure, *next_alt* is called to clear the failed path; if the resulting cleaned tree exists, *runT* is recursively called on it.

$$\frac{\text{success A} \qquad \text{get\_subst } s_1 \text{ A} = s_2 \qquad \text{next\_alt true A} = \text{B}}{\text{runT } fv_0\ s_1\ \text{A}\ s_2\ \text{B}}\ \text{RUN\_DONE}$$

$$\frac{\text{path\_atom A} \qquad \text{step u p } fv_0\ s_1\ \text{A} = (fv_1, \text{st, B}) \qquad \text{runT } fv_1\ s_1\ \text{B}\ s_2\ \text{r}}{\text{runT } fv_0\ s_1\ \text{A}\ s_2\ \text{r}}\ \text{RUN\_STEP}$$

$$\frac{\text{failed A} \qquad \text{next\_alt false A} = \text{Some B} \qquad \text{runT } fv_0\ s_1\ \text{B}\ s_2\ \text{r}}{\text{runT } fv_0\ s_1\ \text{A}\ s_2\ \text{r}}\ \text{RUN\_FAIL}$$

Figure 7 Rule system for *runT*

5 Elpi semantics

We now want to introduce the elpi semantics. The interpreter we show reflects the interpreter of the Elpi language and is an operational semantics close to the one picked by Pusch in [16], in turn closely related to the one given by Debray and Mishra in [6, Section 4.3]. Pusch mechanized the semantics in Isabelle/HOL together with some optimizations that are present in the Warren Abstract Machine [20, 1].

The inductive representing a state of the Elpi language is shown below.

```
Inductive alts := no_alt | more_alt of (Σ * goals) & alts
with goals := no_goals | more_goals of (A * alts) & goals .
```

An elpi state is an enhanced two-dimension list. The outermost list represents the list of alternatives in disjunction accompagnate with the substitution that should be used to for their interpretation. The innermost list is a list of atom, representing a list of goals in cunjunctions. These goals are decorated with a pointer to an elpi state, and are used to keep trace of the alternatives that should be kept when a cut is interpreted. We call these, special, alternatives the cut-to alternatives.

The idea of the Elpi interpreter is to receive a list of alternatives. The first alternative consists of a list of goals. Four cases must be taken into account; they are shown in Figure 8. In order to simplify goal retrieval, we split the head of the alternatives from the tail, so that it can be immediately matched in the inductive definition. Note that an empty list of alternatives represents, by definition, a failing elpi state. If the goal list is empty (StopE), then we have, by definition, a success, and the input solution together with the list of alternatives is returned. If the goal list starts with a cut (CutE), then the current alternatives are erased in favour of the cut alternatives, and a recursive call is made on the remaining goal list.

```
Definition stepE fv t s a gl :=
  let (fv', rs) := bc u p fv t s  in
  let rs_ca := save_alts a gl (r2a rs) in
  (fv', rs_ca).
```

```
Inductive runE : F_v → Σ → goals →  alts → Σ → alts → Prop :=                (10)
```

$$\frac{}{\text{runE fv s [::] a s a}}\ \textsc{StopE}$$

$$\frac{\text{runE fv s gl ca } s_1 \text{ r}}{\text{runE fv s [:: (cut, ca) \& gl] a } s_1 \text{ r}}\ \textsc{CutE}$$

$$\frac{\text{stepE fv t s al gl} = \text{(fv', [:: b \& bs])} \qquad \text{runE fv' b.1 b.2 (bs++al) } s_1 \text{ r}}{\text{runE fv s [:: (call t, ca) \& gl] al } s_1 \text{ r}}\ \textsc{CallE}$$

$$\frac{\text{stepE fv t s al gl} = \text{(fv', [::])} \qquad \text{runE fv' } s_1 \text{ a al } s_2 \text{ r}}{\text{runE fv s [:: (call t, ca) \& gl] [:: } (s_1, \text{a}) \text{ \& al] } s_2 \text{ r}}\ \textsc{FailE}$$

Figure 8 Rule system for *runE*

Finally, we must consider the case in which the goal list starts with a call. The call can either fail (FailE) or succeed (CallE). We distinguish the two cases by looking if the backchaining operation returns zero or more rules. We have wrapped this task in the *stepE* procedure, which also updates the goal and cut-alternative list. The fail case, is relatively easy: the first goal does not succeed, we need to take the head of the alternatives, and make it the new list of goals to be explored.

The case in which backchaining produces a non empty list, the *save_alts* routine is in cahrge of: taking the list of premises and add to each atom the the list of alternatives a as their new cut-alternatives, then it append the list of goals gl to each of these new lists.

6 Semantic equivalence

The equivalence between the two semantics is possible under two conditions: we need to work with "valid tree", i.e. all of those trees that can be generated from a call. Secondly, we need to translate trees into elpi states. In the next to subsection we propose the two functions *valid_state* and *t2l*.

6.1 Valid trees

The inductive tree allows one to generate a large number of trees, some of which are not valid, in the sense that they cannot be produced starting from a given query. The class of valid trees is characterized by the function shown in Figure 9a.

While all base cases of tree are considered valid, we need to analyze carefully the cases for the *Or* and *And* constructors.

For the *Or* constructor, we distinguish two cases depending on whether the left-hand side (lhs) exists. If it does not exist, then the right-hand side (rhs) must be a valid tree. Otherwise, the lhs must itself be a valid tree, and the rhs is either the *KO* tree, since it may have been removed by the evaluation of a superficial cut in the lhs, or it has not yet been

```
Fixpoint valid_tree A :=
  match A with
  | TA _ | OK | KO ⇒ true
  | Or None _ B ⇒ valid_tree B
  | Or (Some A) _ B ⇒ valid_tree A &&
       ((B == KO) || B.base_or B)
  | And A B0 B ⇒ valid_tree A &&
      if success A then valid_tree B
      else B == big_and B0
  end.
```

(a) Valid tree

```
Fixpoint t2l A s0 bt :=
  match A with
  | OK           ⇒ [:: (s0, [::])]
  | KO           ⇒ [::]
  | TA a         ⇒ [:: (s0, [:: (a, [::]) ])]
  | Or None s1 B ⇒ add_ca_deep bt (t2l B s1 [::])
  | Or (Some A) s1 B    ⇒
      let lB := t2l B s1 [::] in
      let lA := t2l A s0 lB in
      add_ca_deep bt (lA ++ lB)
  | And A B0 B   ⇒
      let lA  := t2l A s0 bt in
      let lB0 := a2g B0 in
      let lA  := add_deep bt lB0 lA in
      if lA is [:: (s0, gs) & al] then
        let al := map (catr lB0) al in
        let lB := t2l B s0 (al ++ bt) in
        map (catl gs) lB ++ al
      else [::]
  end.
```

(b) Tree to list

Figure 9 Valid tree and Tree to list

explored. In the latter case, it is a `base_or` tree, namely the right-skewed tree formed by a disjunction of conjunctions that is genereated after a succeful backchain to a call.

For the *And* constructor, the lhs is required to be a valid tree. The shape of the rhs depends on whether the lhs is a success. If the lhs is not successful, then the rhs has never been explored: the procedures *step* and *next_alt* modify the rhs only when the lhs succeeds. In this case, the lhs must be the right-skewed tree containing conjunctions of premises atoms, the reset point B_0 ensure what shape the rhs should have. If the lhs is a success tree, then the rhs can be modified by *step* and *next_alt*, therefore it must be a valid tree.

6.2 From trees to lists

The translation of a tree to a list is shown in Figure 9b. It takes the tree to be translated, a substitution (called s), a list of alternatives, called *bt*. The substitution s tells what is the subsitution of the alternative we are building and is updated when going to the rhs of the *Or* constructor. The *bt* list represents the future alternatives list that have the same depth of the current tree root. They are useful to know if they should be added to the current goal as cut alternatives.

An *OK* node represent a success in a tree, then the translation of this tree returns a singleton list with the cuple $(s, [::])$: there is one alternative with 0 goals, i.e. a success in elpi by StopE. The *KO* node represent a failure, and, therefore 0 alternatives. An atom is translated into a singleton with the atom as first goal and the empty cut-alternative list: we are not adding *bt* since they are the alternatives for the right-sibling, not the right-uncles, this means that if we have *a* is a cut, then it erases the alternatives bt.

In a disjunction, we are scendendo di un livello the distance from our backtracking list. This means that *bt* becomes the list of right-uncles of the two branches of the *Or* constructor. The compilation of the rhs is done independently of if the lhs exists: we transform the rhs

into an elpi state and passing the empty list of alternatives, since the rhs as no right-siblings. Then, thanks to `add_ca_deep`, we recursively add bt to every cut-alternative appearing in the translated goals. If the lhs exists, we translate it and pass the translation of the rhs as the list of right-siblings. Then we prepend the translated lhs to the translated rhs and add bt with `add_ca_deep`.

We compile the *And* case by translating its lhs to the list lA and reset point to the list $lB0$. Then, thanks to `add_deep`, we recrusively append $lB0$ to the alternatives born in lA, i.e. we leave unchanged the pointers to bt, if any, in lA. We finally need to treat cases whenever the list lA is empty or not. In the former case the empty list is returned, otherwise, we hasve a list $[:: (s_0, gs)\&al]$. By the semantics of the *And* in the $runT$ interpreter, the rhs of the *And* represent the sequent of goals to be executed after the first alternative in the lhs, it corresponds to the list of goals gs. The reset point $B0$ is to be appended to the next alternatives in the lhs, that is the queue of lA, that is al.

alternative in tree? rephrase

We note therefore that the compilation of lB is done by passing the alternatives $al + +bt$ since the alternatives in al are to be...

6.3 Equivalence theorems

We have now all the ingredients that ...

```
Lemma tree_to_elpi: ∀ u p s0 t s2 t',
  let fv := vars_tree t `|` vars_sigma s0 in
  valid_tree t →
    runT u p fv s0 t s2 t' →
      ∃ s1 g a,
        let: na := t2l (odflt KO t') s0 [::] in
        t2l t s0 [::] = (s1,g) :: a /\
          runE u p fv s1 g a s2 na.
```

(5)

```
Lemma elpi_to_tree: ∀ u p fv s1 g s2 a na,
  runE u p fv s1 g a s2 na →
    ∀ s0 t, valid_tree t → t2l t s0 [::] = (s1,g) :: a →
      ∃ t', runT u p fv s0 t s2 t' /\ t2l (odflt KO t') s0 [::] = na.
```

(6)

The proof of Lemma 6 is based on the idea explained in [2, Section 3.3]. An ideal statement for this lemma would be: given a function `l2t` transforming an elpi state to a tree, we would have have that the the execution of an elpi state e is the same as executing $runT$ on the tree resulting from `l2t`(e). However, it is difficult to retrive the strucutre of an elpi state and create a tree from it. This is because, in an elpi state, we have no clear information about the scope of an atom inside the list and, therefore, no evident clue about where this atom should be place in the tree.

Our theorem states that, starting from a valid tree t which translates to a list of alternatives $(\sigma_1, g) :: a$. If we run in elpi the list of alternatives, then the execution of the tree t returns the same result as the execution in elpi. The proof is performed by induction on the derivations of the elpi execution. We have 4 derivations.

We have 4 case to analyse:

7 Case study: determinacy alanysis

we mechanize the first order part of xxx.

snippet det, main thm, invariant det tree (valid tree prev section?)

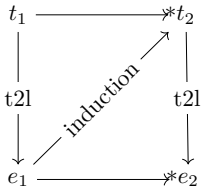


Figure 10 Induction scheme for Lemma 6

proof induciton on exec, step/next alt preserving invariant proved by induction on the tree.

with list semantics cut and next alt requires to express a link btween the ca or next alts and the current goal, which is non trivial without an intermediate data strature like the tree

8 Related work

prolog semantics, King lost

yves for the proof technique

9 Conclusion

References

1. Hassan Aït-Kaci. *Warren's Abstract Machine: A Tutorial Reconstruction*. The MIT Press, 08 1991. doi:10.7551/mitpress/7160.001.0001.
2. Yves Bertot. A certified compiler for an imperative language. Technical Report RR-3488, INRIA, September 1998. URL: https://inria.hal.science/inria-00073199v1.
3. Valentin Blot, Denis Cousineau, Enzo Crance, Louise Dubois de Prisque, Chantal Keller, Assia Mahboubi, and Pierre Vial. Compositional pre-processing for automated reasoning in dependent type theory. In Robbert Krebbers, Dmitriy Traytel, Brigitte Pientka, and Steve Zdancewic, editors, *Proceedings of the 12th ACM SIGPLAN International Conference on Certified Programs and Proofs, CPP 2023, Boston, MA, USA, January 16-17, 2023*, pages 63–77. ACM, 2023. doi:10.1145/3573105.3575676.
4. Cyril Cohen, Enzo Crance, and Assia Mahboubi. Trocq: Proof transfer for free, with or without univalence. In Stephanie Weirich, editor, *Programming Languages and Systems*, pages 239–268, Cham, 2024. Springer Nature Switzerland.
5. Cyril Cohen, Kazuhiko Sakaguchi, and Enrico Tassi. Hierarchy Builder: Algebraic hierarchies Made Easy in Coq with Elpi. In *Proceedings of FSCD*, volume 167 of *LIPIcs*, pages 34:1–34:21, 2020. URL: https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.FSCD.2020.34, doi:10.4230/LIPIcs.FSCD.2020.34.
6. Saumya K. Debray and Prateek Mishra. Denotational and operational semantics for prolog. *J. Log. Program.*, 5(1):61–91, March 1988. doi:10.1016/0743-1066(88)90007-6.
7. Cvetan Dunchev, Ferruccio Guidi, Claudio Sacerdoti Coen, and Enrico Tassi. ELPI: fast, embeddable, λProlog interpreter. In *Proceedings of LPAR*, volume 9450 of *LNCS*, pages 460–468. Springer, 2015. URL: https://inria.hal.science/hal-01176856v1, doi:10.1007/978-3-662-48899-7_32.
8. Davide Fissore and Enrico Tassi. A new Type-Class solver for Coq in Elpi. In *The Coq Workshop*, July 2023. URL: https://inria.hal.science/hal-04467855.

9 Davide Fissore and Enrico Tassi. Higher-order unification for free!: Reusing the meta-language unification for the object language. In *Proceedings of PPDP*, pages 1–13. ACM, 2024. doi:10.1145/3678232.3678233.

10 Davide Fissore and Enrico Tassi. Determinacy checking for elpi: an higher-order logic programming language with cut. In *Practical Aspects of Declarative Languages: 28th International Symposium, PADL 2026, Rennes, France, January 12–13, 2026, Proceedings*, pages 77–95, Berlin, Heidelberg, 2026. Springer-Verlag. doi:10.1007/978-3-032-15981-6_5.

11 Benjamin Grégoire, Jean-Christophe Léchenet, and Enrico Tassi. Practical and sound equality tests, automatically. In *Proceedings of CPP*, page 167–181. Association for Computing Machinery, 2023. doi:10.1145/3573105.3575683.

12 Ferruccio Guidi, Claudio Sacerdoti Coen, and Enrico Tassi. Implementing type theory in higher order constraint logic programming. In *Mathematical Structures in Computer Science*, volume 29, pages 1125–1150. Cambridge University Press, 2019. doi:10.1017/S0960129518000427.

13 Robbert Krebbers, Luko van der Maas, and Enrico Tassi. Inductive Predicates via Least Fixpoints in Higher-Order Separation Logic. In Yannick Forster and Chantal Keller, editors, *16th International Conference on Interactive Theorem Proving (ITP 2025)*, volume 352 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 27:1–27:21, Dagstuhl, Germany, 2025. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. URL: https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.ITP.2025.27, doi:10.4230/LIPIcs.ITP.2025.27.

14 Dale Miller. A logic programming language with lambda-abstraction, function variables, and simple unification. In *Extensions of Logic Programming*, pages 253–281. Springer, 1991.

15 Dale Miller and Gopalan Nadathur. *Programming with Higher-Order Logic*. Cambridge University Press, 2012.

16 Cornelia Pusch. Verification of compiler correctness for the wam. In Gerhard Goos, Juris Hartmanis, Jan van Leeuwen, Joakim von Wright, Jim Grundy, and John Harrison, editors, *Theorem Proving in Higher Order Logics*, pages 347–361, Berlin, Heidelberg, 1996. Springer Berlin Heidelberg.

17 Enrico Tassi. Elpi: an extension language for Coq (Metaprogramming Coq in the Elpi λProlog dialect). In *The Fourth International Workshop on Coq for Programming Languages*, January 2018. URL: https://inria.hal.science/hal-01637063.

18 Enrico Tassi. Deriving proved equality tests in Coq-Elpi. In *Proceedings of ITP*, volume 141 of *LIPIcs*, pages 29:1–29:18, September 2019. URL: https://inria.hal.science/hal-01897468, doi:10.4230/LIPIcs.CVIT.2016.23.

19 Luko van der Maas. Extending the Iris Proof Mode with inductive predicates using Elpi. Master's thesis, Radboud University Nijmegen, 2024. doi:10.5281/zenodo.12568604.

20 David H.D. Warren. An Abstract Prolog Instruction Set. Technical Report Technical Note 309, SRI International, Artificial Intelligence Center, Computer Science and Technology Division, Menlo Park, CA, USA, October 1983. URL: https://www.sri.com/wp-content/uploads/2021/12/641.pdf.